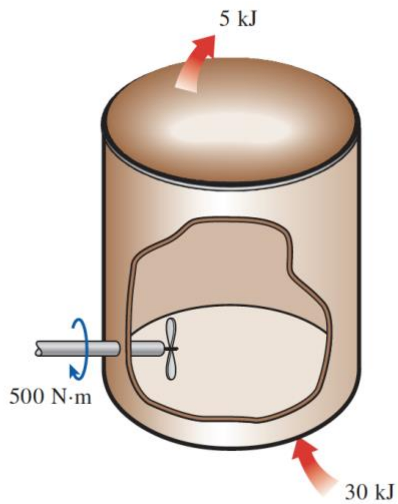
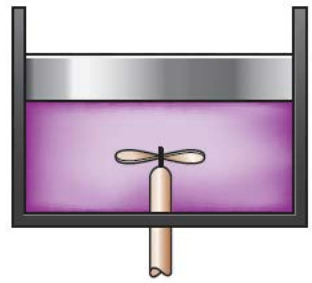
# Chapter 2 Workshop

1. Water is being heated in a closed pan on top of a range while being stirred by a paddle wheel. During the process, 30 kJ of heat is transferred to the water, and 5 kJ of heat is lost to the surrounding air. The paddle-wheel work amounts to 500 N.m. Determine the final energy of the system if its initial energy is 12.5 kJ.



2. Consider a 2100-kg car cruising at constant speed of 70 km/h. Now the car starts to pass another car by accelerating to 110 km/h in 5 s. Determine the additional power needed to achieve this acceleration. What would your answer be if the total mass of the car were only 700 kg? (The power output is due to kinetic energy only).

3. A closed system is operated in an adiabatic manner. First, 15,000 lbf·ft of work are done by this system. Then, work is applied by the stirring device to the system in the amount of 10.28 Btu. Calculate the net increase in the internal energy of this system.



4. A fan is to accelerate quiescent air to a velocity of 8 m/s at a rate of 9 $m^3$/s. Determine the minimum power that must be supplied to the fan. Take the density of air to be 1.18 kg/$m^3$. The power output is due to kinetic energy only.